

SICILIAN AIRLINES



Passenger Satisfaction

A Data Science Case Study

Predicting what makes passengers happy — and how to act on it.

Dataset

103,904 passengers

Task

Binary Classification

Models

**Logistic Regression · Random Forest ·
AdaBoost · Gradient Boosting**

Baseline → Best Score

~57% → ~97% Accuracy (+40pp vs baseline)

Who We Are

Sicilian Airlines

We are an airline with a clear double mission:
deliver an exceptional passenger experience and minimise our environmental footprint at every stage of travel.

Why this project?

Understanding what drives passenger satisfaction is the starting point for smarter decisions — from cabin service to ground operations — without wasting resources on the wrong improvements.



Air Excellence

Premium in-flight experience



Eco Commitment

Reduce environmental impact



Passenger First

Data-driven decisions

The Business Problem

Can we predict whether a passenger will be satisfied — and identify which service dimensions drive that outcome?

**103,904
passengers**

High-volume dataset requiring robust preprocessing

**24
features**

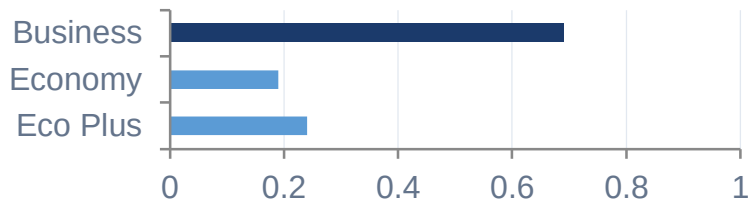
Mix of ratings, demographics and flight metadata

**Binary
target**

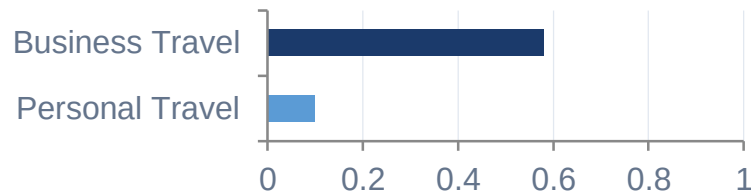
Satisfied vs. Neutral/Dissatisfied — classification task

Exploratory Analysis — Key Insights

Satisfaction Rate by Class



Satisfaction Rate by Travel Type



Business class passengers are 3–4× more likely to be satisfied than Economy.



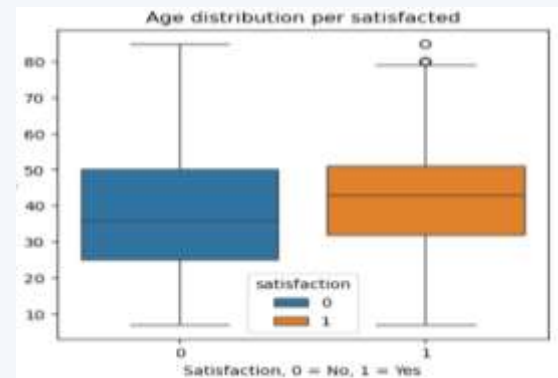
Business travelers show ~58% satisfaction vs. ~10% for personal/leisure travelers.



Online boarding, Inflight entertainment and Seat comfort show the largest gap between satisfied and dissatisfied passengers.



Satisfied passengers tend to be slightly older — median age ~43 vs ~36 for dissatisfied.



Dataset & Preprocessing

01

Load & Inspect

103,904 rows · 25 columns. Only 310 missing values in 'Arrival Delay' (~0.3%) — filled with median.

02

Drop Redundant Columns

"id" (pure identifier) and "Arrival Delay in Minutes" removed — correlation with Departure Delay ~0.96.

03

Encode Target

satisfaction → 0 (neutral/dissatisfied) · 1 (satisfied). Distribution: 57% vs 43% — slightly imbalanced.

04

Train / Val / Test Split

70% train · 15% validation · 15% test. Cascaded split to keep all three sets fully disjoint.

05

Preprocessing Pipeline

Numerical: median imputation. Categorical (Gender, Class, Travel Type): OrdinalEncoder. Wrapped in sklearn Pipeline to prevent leakage.

06

Outlier Analysis

Flight Distance and Departure Delay show long right tails (IQR confirms outliers). Not removed — they carry real signal. Median imputation chosen for robustness.

Choosing the Right Metrics

Accuracy alone is not enough. Given the slight class imbalance, we track four complementary metrics:

Accuracy

% of total predictions that are correct.
Baseline — easy to interpret but can mislead on imbalanced data.

Precision

Of all passengers predicted satisfied,
how many actually were? Avoids false positives.

Recall

Of all truly satisfied passengers,
how many did the model catch? Avoids false negatives.

F1-Score

Harmonic mean of Precision & Recall.
Best single metric when both matter equally.

Which Features Matter Most?

Three complementary statistical tests were run on the training set. Results are consistent across methods.

Chi-Square

Statistical dependency between each feature and the target.

All of these have a p-value practically equal to zero, so the probability that their association with the target variable is due to chance is virtually nonexistent.

Type of Travel

Class

Flight Distance

Age

Customer Type

Mutual Information

Information each feature provides about the target (captures non-linear links).

Online boarding

Inflight wifi service

Class

Type of Travel

Inflight entertainment

T-Test (Welch)

Mean difference between satisfied and dissatisfied groups. 4 features selected ($p < 0.001$).

Age

Flight Distance

Inflight wifi service

Ease of Online booking

Modeling Strategy

We explore two different approaches to see which produces the best results.

Road 1

All available features (22 columns after cleaning)

1

Spot Check

Stratified 5-Fold CV with default hyperparameters — compare Random Forest, AdaBoost, Gradient Boosting. Scoring: Accuracy, F1, ROC-AUC.

3

Hyperparameter Tuning

GridSearchCV (5-fold) explores `n_estimators`, `max_depth`, `learning_rate`, `subsample` and leaf constraints.

Road 2

Top 4 features from T-test only

2

Select Top 2

Pick the two best-performing models based on mean CV accuracy. Proceed to tuning only those.

4

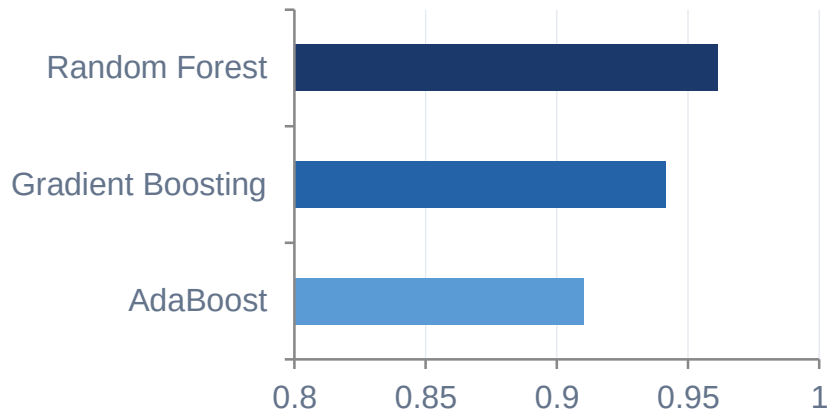
Final Evaluation

Best models evaluated on the held-out test set — untouched until this moment.

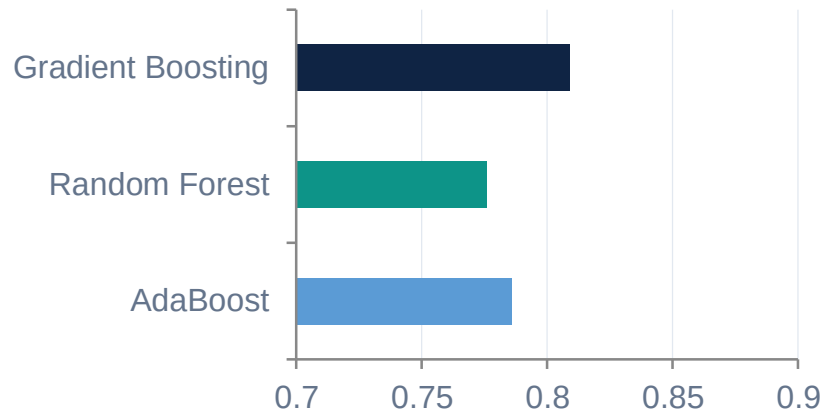
Spot Check — 5-Fold Cross Validation

Default hyperparameters · Stratified KFold $k=5$ · Scoring: Accuracy, F1, ROC-AUC

Road 1 — All Features



Road 2 — Top 4 Features



Takeaway: Random Forest leads on Road 1 · Gradient Boosting leads on Road 2 → both advance to hyperparameter tuning.

Hyperparameter Tuning — GridSearchCV

5-fold cross validation on training set. Test set untouched.

Random Forest · Road 1

n_estimators [100, **200**, 300]

max_depth [**None**, 10, 20]

min_samples_split [**2**, 5, 10]

min_samples_leaf [**1**, 2, 4]

Best CV score: ~96.3% accuracy

Using the optimal parameters, performance improves from 96% to 97%.

Gradient Boosting · Road 2

n_estimators [100, 200, **300**]

learning_rate [0.05, 0.1, **0.2**]

subsample [**0.8**, 1.0]

min_samples_leaf [1, 2, **4**]

Best CV score: ~81.3% accuracy

Using the optimal parameters, performance improves from 77% to 81%.

Final Results — Test Set Evaluation

ROAD 1 — All Features

Random Forest (tuned)

Accuracy ~97%

Precision ~97%

Recall ~97%

F1-Score ~97%

ROAD 2 — Top 4 Features

Gradient Boosting (tuned)

Accuracy ~82%

Precision ~82%

Recall ~81%

F1-Score ~81%

Conclusions

Best Model:

Random Forest with GridSearchCV tuning — ~97% accuracy on unseen data.

Key Drivers:

Class, Type of Travel, Online boarding and Inflight wifi are the strongest predictors of satisfaction.

Feature Efficiency:

Road 2 (4 features) reaches ~82% — good, but 22 features is better. More data wins.

Action Points:

- Fix the online boarding experience — largest gap across all passengers.
- Invest in inflight entertainment and seat comfort — second and third largest gaps.
- Upgrade inflight wifi — top predictor across all three feature selection methods.
- Improve the Economy cabin — satisfaction rate is less than 20%, the biggest untapped segment

Limitations:

Single-airline snapshot · Self-reported rating bias ·
OrdinalEncoder assumes uniform distance
between cabin classes.

Next Steps:

Try XGBoost/LightGBM, analyse feature importances, explore
SMOTE for class balance. Run seed analysis across multiple splits
to confirm result stability.



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